HTFE SOLUTIONS



R&D Consulting / Engineering Simulation Services

- Innovative Solutions
- Client-Centric Approach
- Technical Proficiency
- Proven Track Record

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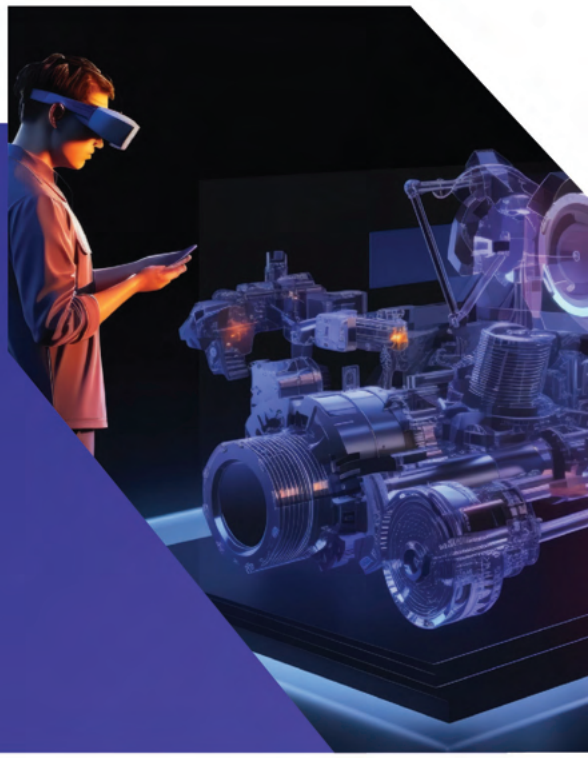
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HTFE SOLUTIONS

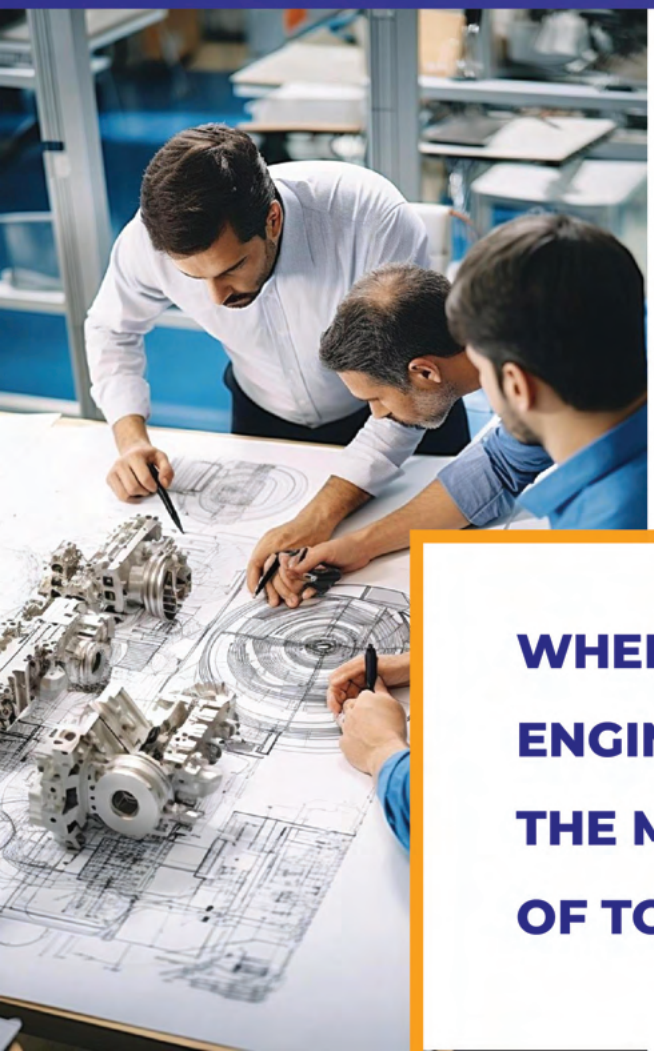
Who are we?



HTFE Solutions is the Design Service provider company established inn 2014.

We are continuously providing Design Services in Mechanical, Electrical and Architecutural Design.

We have the team of Expertise Designers in their respective CAD Software.



WELCOME TO HTFE

WHERE INNOVATION MEETS ENGINEERING EXCELLENCE TO SOLVE THE MOST INTRICATE CHALLENGES OF TODAY AND TOMORROW



Expertise

- Multiphysics analysis
- Multiphase analysis
- Non-linear analysis
- Transient dynamic analysis
- Multibody simulation
- Vibration a.
- Seismic a
- System Modelling & Analysis
- Conjugate Heat Transfer
- Non-Newtonian Flows
- Compressible flows
- FSI
- Structural analysis
- Fire & smoke simulation
- Fluid mixture, separation, particulate
- Particle flow
- Crash Worthiness
- Fatigue, buckling analysis
- Modal analysis
- Aero/hydroacoustic
- Acoustic analysis
- Solid Modelling
- Revit BIM
- HVAC Design & Development
- Product Design & Development
- Process Engineering
- Piping Analysis

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1.1

CASE STUDIES
AUTOMOBILE - INDUSTRY
Automobile Engine - Air Intake System

Title

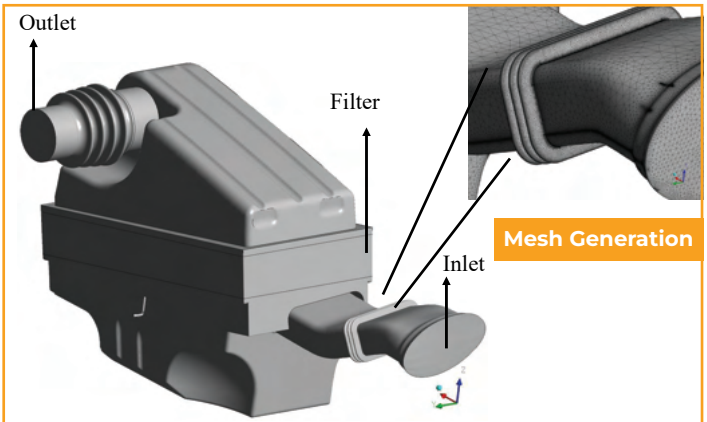
CFD - Pressure Drop Analysis of Automobile Engine Air Intake System.

Objective

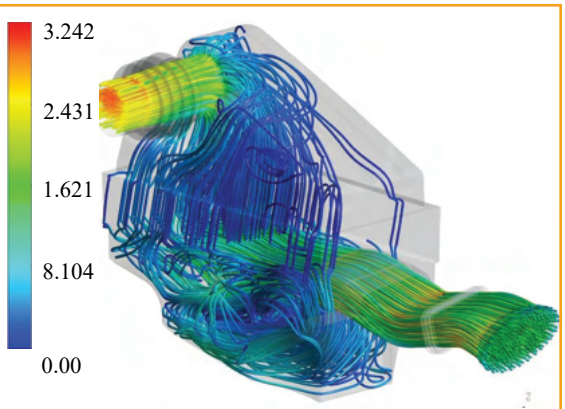
Perform CFD analysis on components of air intake assembly to predict the pressure drop for given mass flow rate and temperature conditions.

Pressure - Drop summary

OUTPUT		
Model	Componenet	Status as per CFD
31	Air DuctInlet + Air Filter Box	PASS

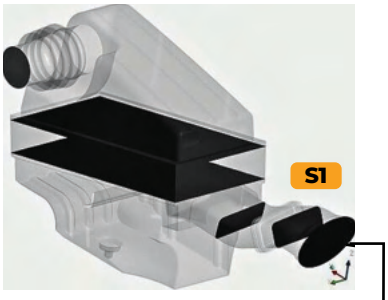


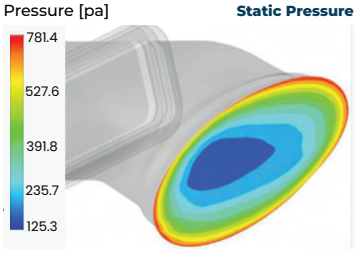
3D Model and mesh of Automobile Engine Air Intake System

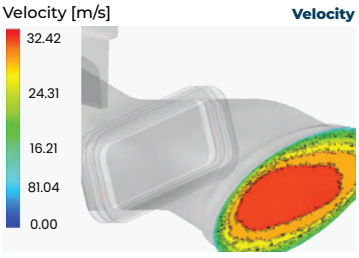


Automobile Engine - Air Intake System Streamlines

Static Pressure & Velocity Contour Plot at S1

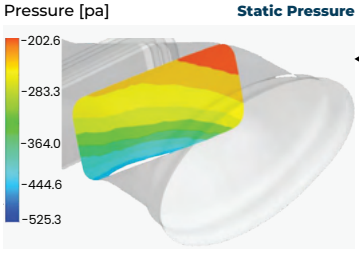


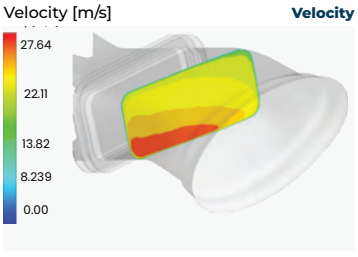




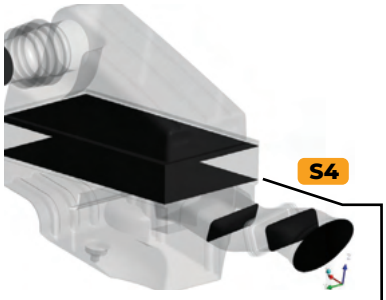
Static Pressure & Velocity Contour Plot at S2

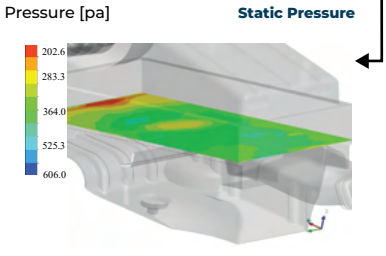


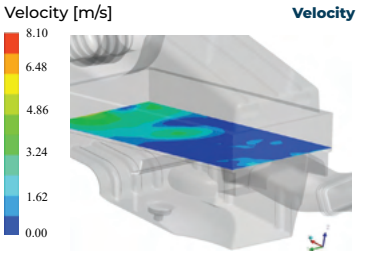




Static Pressure & Velocity Contour Plot at S4







1.2

CASE STUDIES
Turbo - Machinery

Title

Automobile Centrifugal Blower **Design & Optimizations** are suggested and implemented throughout this study.

Objective

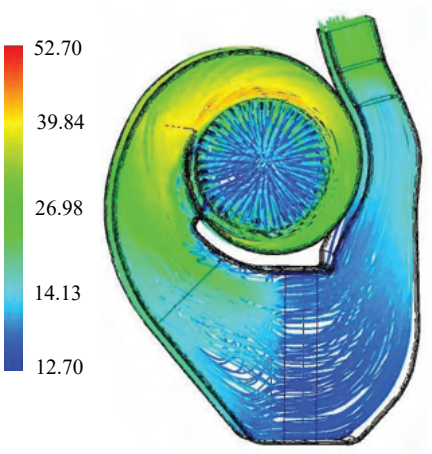
3D Simulation of Automobile air conditioning system centrifugal blower to:

- Predict **flow rate** & distribution through HVAC passages.
- Suggest **design optimizations.**

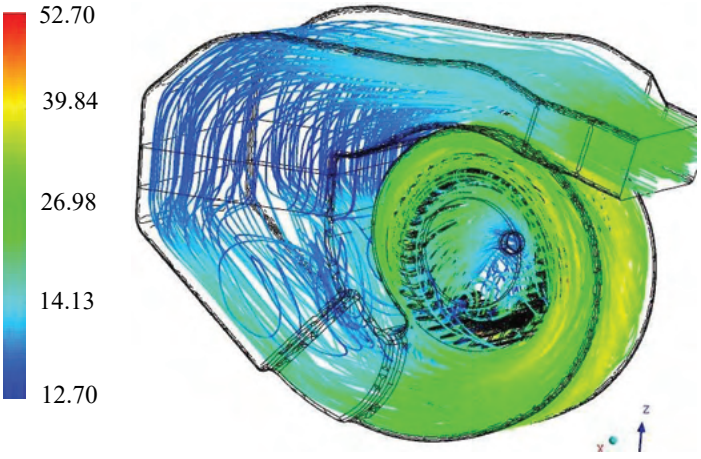
Output

- **Tapering** can be provided in this portion to avoid sudden change in flow path to avoid recirculation observed in the diverging duct just before heater.
- **Baffles** can be provided along the whole cross section to guide the flow to reduce swirl and recirculation in this portion.
- **Radius of curvature** can be increased at this portion of duct. This will better assist the flow and reduce the cross sectional area to slightly improve flow rate.

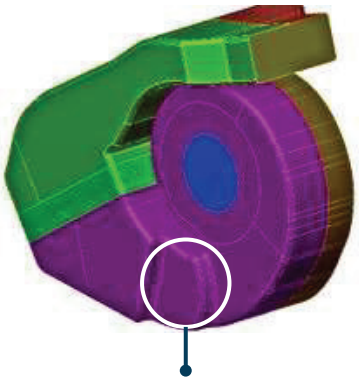
Velocity [m/s]
(Streamline inlet)



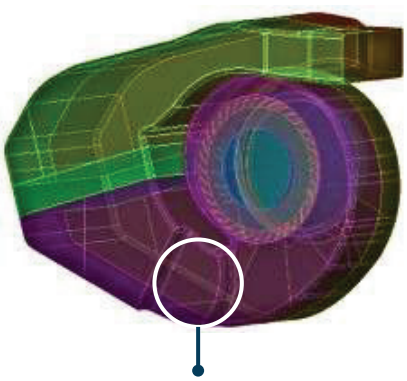
Velocity [m/s]
(Streamline inlet)



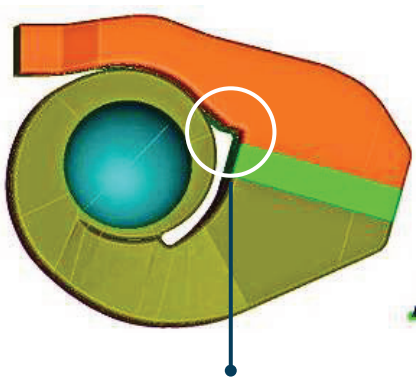
Design Changes Suggestions



Tapering can be provided in this portion to avoid sudden change in flow path to **avoid recirculation observed in the diverging duct** just before heater.



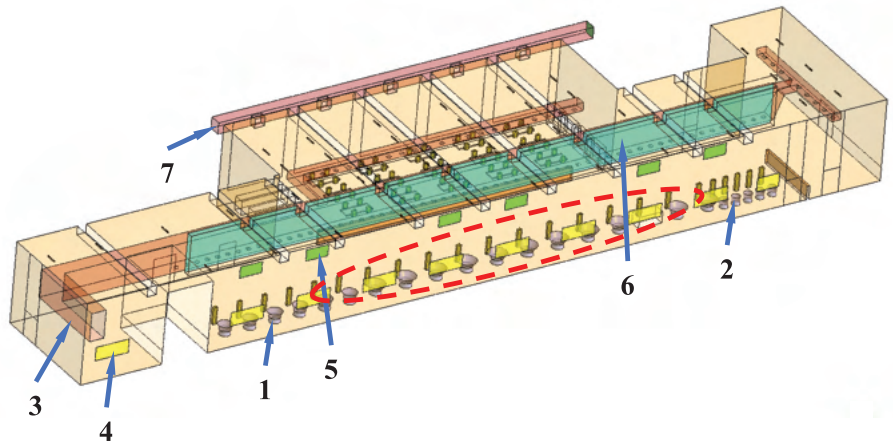
Baffles can be provided along the whole cross section to guide the flow to **reduce swirl and recirculation** in this portion



Radius of curvature can be increased at this portion of duct. This will better **assist the flow and reduce the cross sectional area** to slightly improve flow rate

2.1

CASE STUDIES
HVAC - R



1. Steam Kettles
2. LPG Bhatti
3. Supply Air Duct
4. Window
5. Exhaust Air Grille
6. Pardi Wall
7. Exhaust Air Duct

Client

- No disclosure Agreement with the Client

Project Highlight

- India's biggest FMCG Plant with 400000 Sq.ft. floor area.

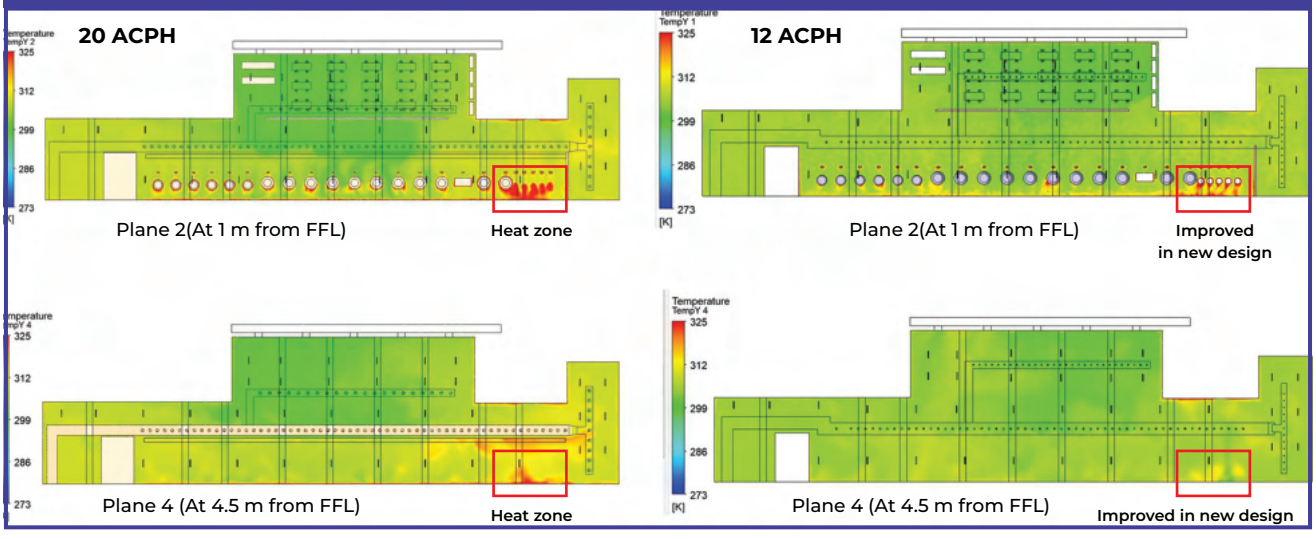
Contribution of HTFE Solutions

- **Validated the consultant design for 20 ACPH**
- **Optimized the design further to achieve the client requirement at 12 ACPH to save the cost as follows -**
  - Up to INR 1.5 CR operational cost of two stage air washer system can be saved per year.
  - Up to INR 1 CR can be saved on ducting supply and installation and other project CAPEX.

Work done by HTFE Solutions

Initially, they have modeled the air-washer system as per consultant's design is at 20 ACPH.	**We performed simulation and optimization in their designs.**
Design is modified to reduce the ACPH value from 20 to 12.	**It can be seen from temperature distribution (find in next slide) is improved remarkably in new design even when shifting from 20 ACPH to 12 ACPH value.**

Temperature Contour Horizontal Plane Comparison



2.2 CASE STUDIES
Horticulture ventilation

Project - CFD Analysis of the Canopy arrangement

Client - Canopy manufacturer in USA

Horticulture ventilation

Project & Scope

Anlayse the existing design.

Optimize the design through CFD analysis.

Challenges

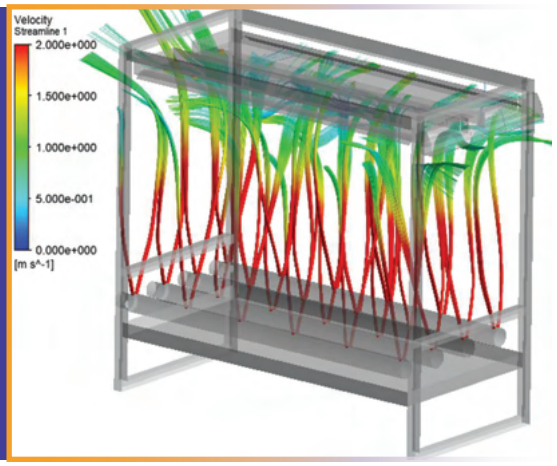
Anlaysis for **unpredictable conditions.**

Maintaining the **uniform flow** in a system.

Optimizing the flow around supply and exhaust duct.

Key Points

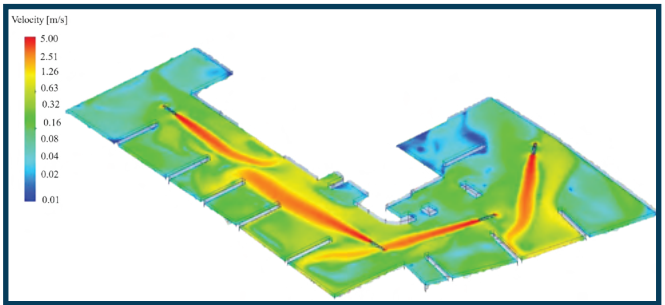
- Analysis was assisted with several rounds of **iterative testing** to qualify the overall progress of the project
- Recommend **optimizing the size and positioning of supply duct inlets, imposing restrictions around exhaust ducts, and adjusting supply CFM**



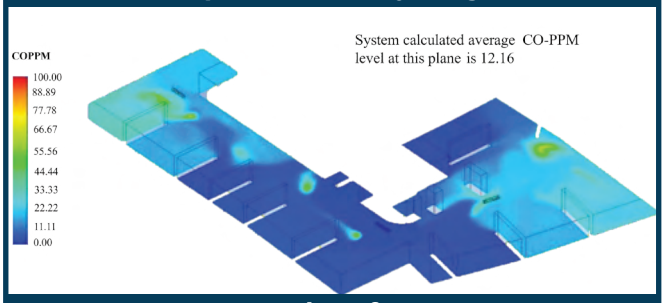
2.3 HVAC - Car-park Ventilation

Objective Of the project

- To check the effectiveness of the ventilations system Forfire, so that the fire and smoke can be put out And there should be **clear visibility** for the people To **escape** at stipulated time.
- The event of a **co high level (>100 ppm)** is avoided At short term exposure of 30 min.
- To investigate the **performance of the system**, so that there is adequate air movement throughout The car park under normal operating conditions.
- The cfd study aims to analyze and simulate all The above mentioned conditions and **validate** The design of the system, in the achievement Of its objectives.



Contourplot of velocity magnitude



Contourplot of co ppm

The mathematical modeling for the analysis involves a few simplifying assumptions and care has been taken to meet the safety requirements. The analysis is done with Jet Fans operating at Normal Mode speed, placed and optimized so as to remove the stagnant areas and to reduce the concentrations of CO in the car park.

2.4

CASE STUDIES
Data Center

Data Center Cooling

Scope

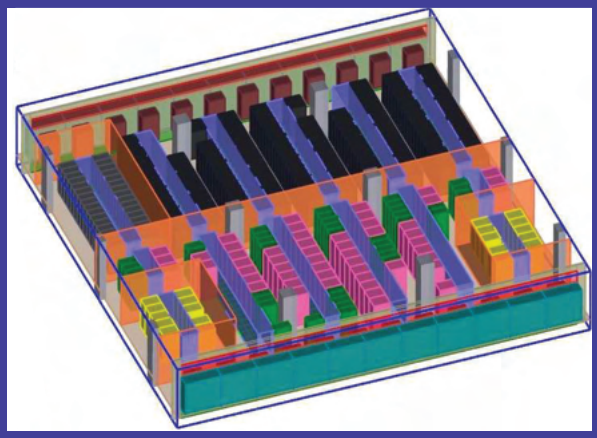
The objective of the project was to check the temperature distribution in & around the racks, short circuiting of cold/hot air and sufficient supply of air to cool the racks. Also the air recirculation zones were checkedto achieve more efficient flow.

Parameters Studied

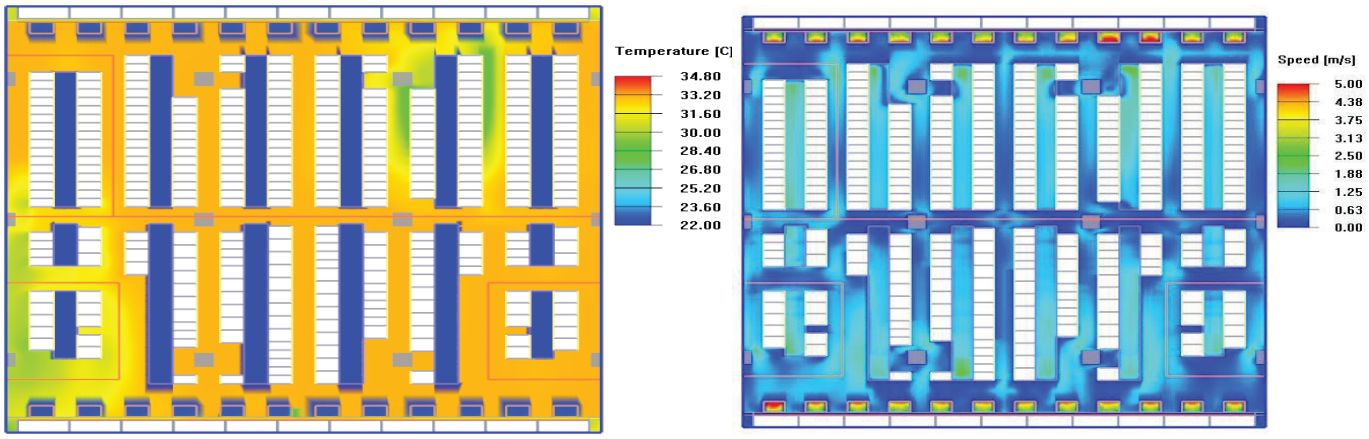
- Velocity distribution
- Temperature profile
- Fresh air supply air rate
- Air recirculation area
- Pressure contour plot in Cold and hot aisle

Report/ Deliverables

- Detail report will be provided.
- Velocity andTemperature distribution at different sections, animations for visual inspections.



- 166 RACKS 990 KW
- 32 RACKS 256 KW
- PDU IN OPERATION
- CAFE WALL
- WIRE MESH
- COLD AISLE CONTAINENT
- 42 RACKS 10 KW EACH
- 53 RACKS 6 KW EACH
- WALL
- COLUMSS
- RAISED FLOOR
- 20 RACKS 10 KW EACH
- PAHU IN OPERATION
- FLOOR GRILLE
- WIRE MESH



3.1

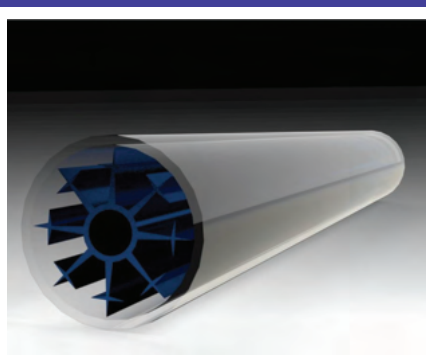
CASE STUDIES • Thermal Analysis

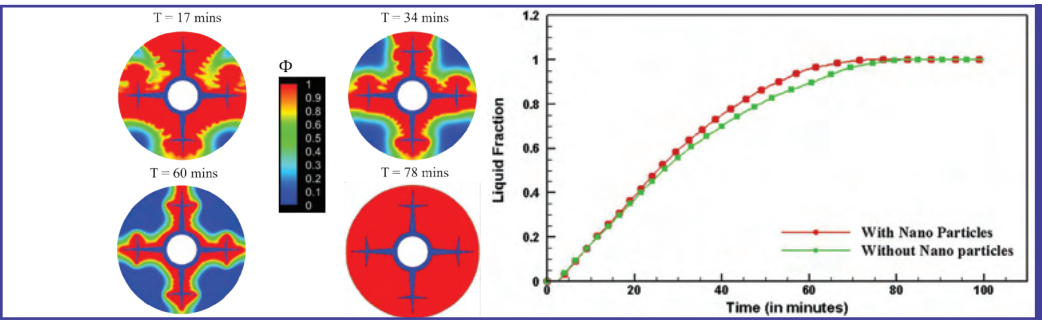
Two Phase modelling of a Latent Heat Storage Unit using Cesaro Fins

Work Done

- Created additional fin types to enhance the system's performance. To investigate the system's performance, transient simulations were also run. As a result, the performance of the system has been studied using a number of combinations, including the type of fins, metal foams, and nanoparticles.
- Conducted a detailed investigation & offered thorough customer support to improve the product. Consequently, a thorough report of the survey has been provided. Different fin combinations, such as Cesaro, cross-type and cross with side hand, were taken into consideration in the study. After picking the best fins, the system was improved by adding different amounts of metal foam and nanoparticles.

- The present problem simulates the process of melting of phase change materials inside a finned tube.
- The simulation is performed in two dimensions and hence only a hollow circular section of the pipe is modelled.
- The phase change material used is RT-82.
- When the temperature around a liquid substance drops, the PCM hardens, and value of Liquid fraction is '0'.
- Whereas value of liquid fraction equal to '1' indicates that PCM is in liquid phase.

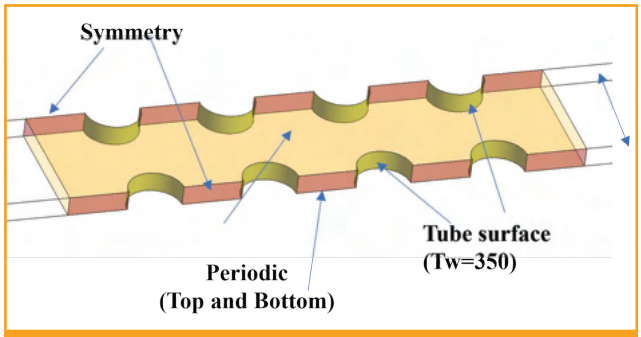




1. The liquid fraction changes faster with time in Nano particle infused PCM material.

2. Copper isused as nano particles with 4% volume fraction

3.2 Thermal Performance Enlargement of Fin-and-Tube Heat Exchanger Using Vortex Generators Including Distinct Circular Punched Holes

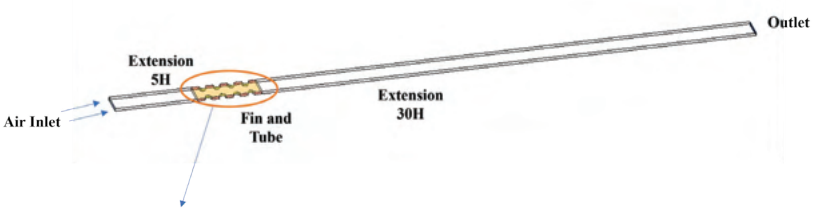


A vortex generator with punched holes

Summary

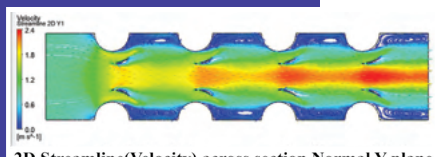
- It has been widely explored to enhance the air-side convective heat transfer coefficient, particularly in plate-fin and tube-fin compact heat exchangers.
- Our customer is a Manufacturer. Exporter and Supplier of various types of heat exchangers.
- A vortex generator with punched holes was studied to explore the performance of the compact heat exchangers.

- HTFE Solutions modelled the vortex generators based on given dimensions and configurations.
- Modelled VGs have been attached to a fin tube heat exchanger so that it increases the heat transfer rate, and the computational domain has been restricted to the region of concern.
- The thermal contact resistance between the tube and fin collar is ignored, and the tube surface is set at a constant temperature.

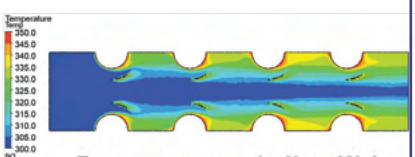


- HTFE Solutions team optimised the design by doing trial & error on VGs locations, orientations and configurations.
- More than 40 simulation data have been compiled and the report has been submitted to the customer for future reference.

Thermal analysis



2D Streamline(Velocity) across section Normal Y plane

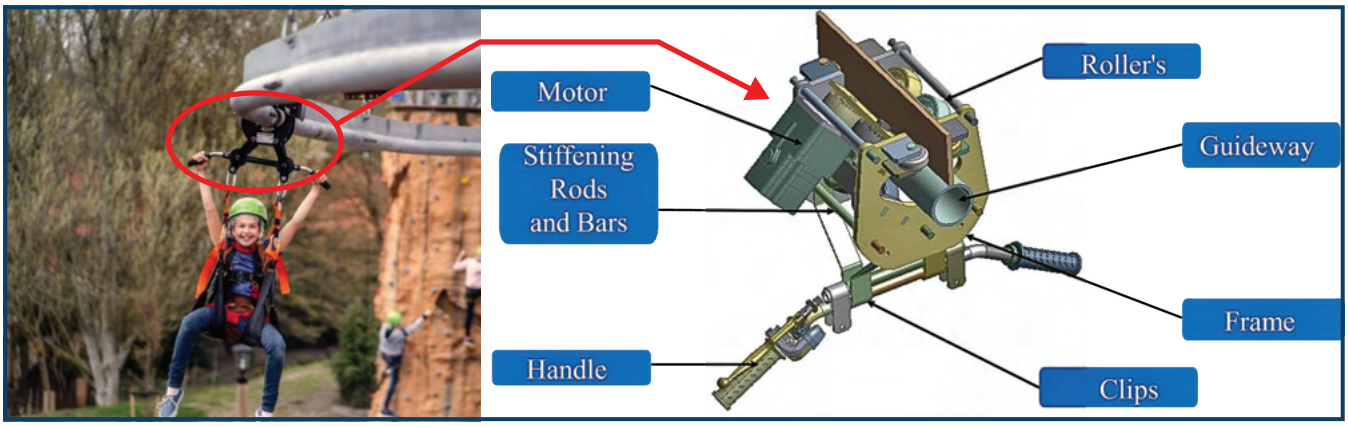


Temperature across section Normal Y plane

4.1

CASE STUDIES - **General Problems**

Finite ElementAnalysis on Roll Glider



Title

- Strength Analysis of Roll Glider

Applicant Requirements

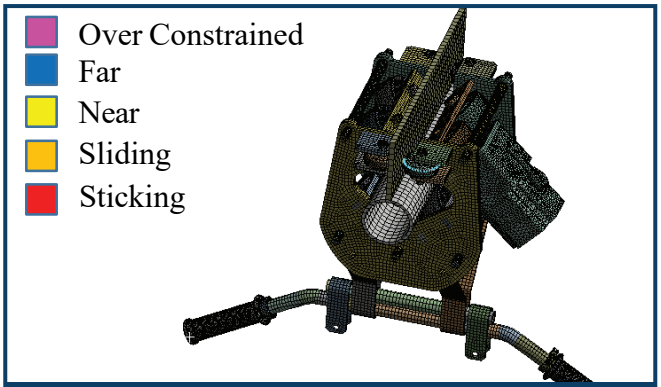
- For the **Roll Glider** unit, it is required to validate the strength of the **Roll Glider assembly** under operational loads.

Scope

- Perform FEA on Frame for following Operational load case.
- LC1 - **Strength Analysis at 500 Kg Load** (linear static)

Output

- The max stress is 242 MPa which is well within the yield limit.
- In order to review the detailed results of roller's while application the dynamic analysis is suggested which will give comprehensive idea about the possible deformations in wheels during motion.
- Considering all above observations for given distributed loading of 500 Kg over anchor and holding locations the design is safe.

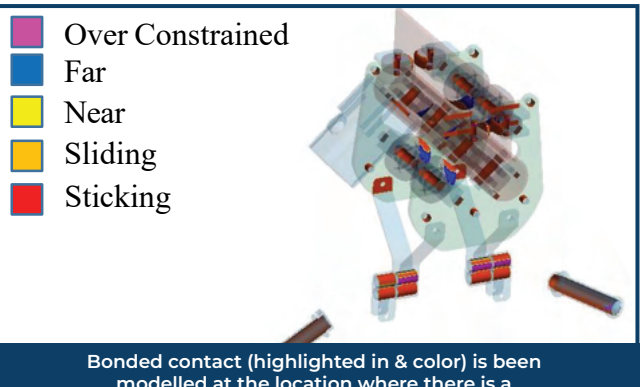


Average element size range is 3 mm

Details of "Mesh"

Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default
Sizing	
Use Adaptive Sizi...	Yes
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Di...	965.02 mm
Average Surface ...	508.64 mm²
Minimum Edge L...	0.2 mm
Quality	
Inflation	
Advanced	
Statistics	
Nodes	550651
Elements	279741

Mesh Setting applied



Bonded contact (highlighted in & color) is been modelled at the location where there is a possibility of fastening of two components

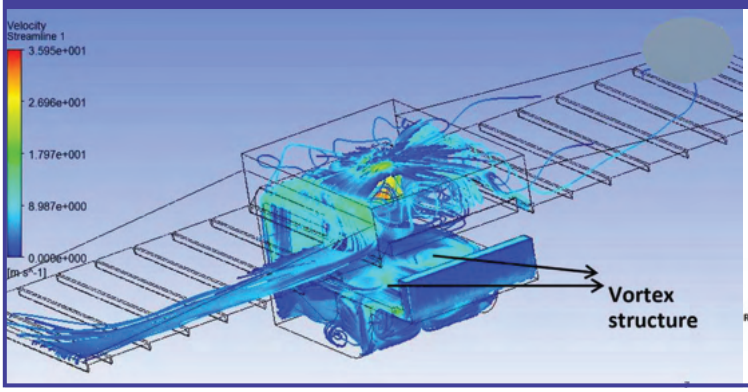
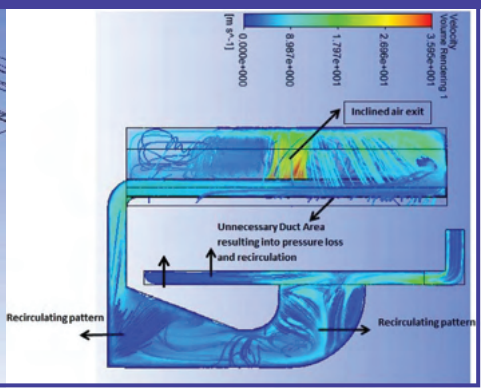
Details of "Bonded - 600441_Standard To 600442.1_"

Scope	
Scoping Method	Geometry Selection
Contact	4 Faces
Target	4 Faces
Contact Bodies	600441_Standard
Target Bodies	600442.1_Standard
Protected	No
Definition	
Type	Bonded
Scope Mode	Automatic
Behavior	Program Controlled
Trim Contact	Program Controlled
Trim Tolerance	2.3327 mm
Suppressed	No
Advanced	
Formulation	MPC
Detection Method	Program Controlled
Constraint Type	Program Controlled
Pinball Region	Radius
Pinball Radius	5. mm
Geometric Modification	
Contact Geometry Correction	None
Target Geometry Correction	None

Contact Setting applied

4.2 CASE STUDIES - General Problems

Designing/optimizing the cloth dryer equipment

Recirculation patterns near inlet region and burner region	Recirculating pattern of air and inclined exit
	

Title

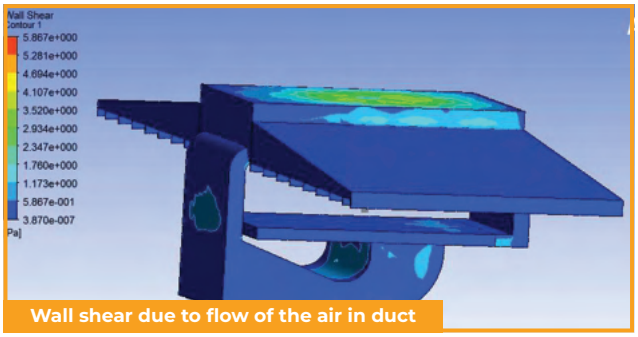
- CFD report on designing/optimizing the cloth dryer equipment

Objective

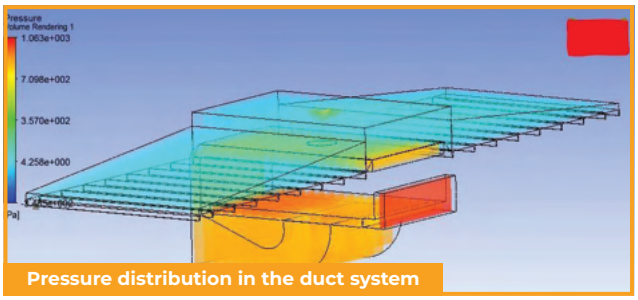
- Design changes based on pressure loss calculations and system resistance to flow of the air.
- Defining the capacity of the fan and its suitability for the present equipment.

Output

- It is observed that equipment design has sharp edges and unnecessary duct areas where the air keeps recirculating resulting into pressure losses.
- To reduce the swirling effect, vortex structure and recirculation of the flow, sharp edges of the ducts should be made round.
- Cross sectional area of the duct can be increased. Also, unnecessary portions of the duct which do not contribute to the structural integrity or smoothening of flow patterns can be eliminated safely.

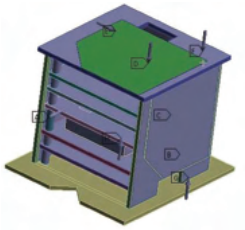


Wall shear due to flow of the air in duct

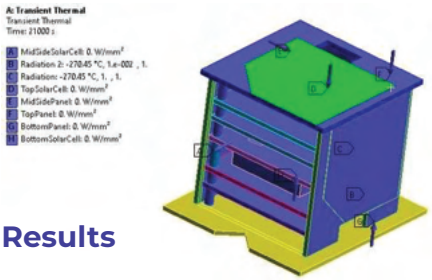


Pressure distribution in the duct system

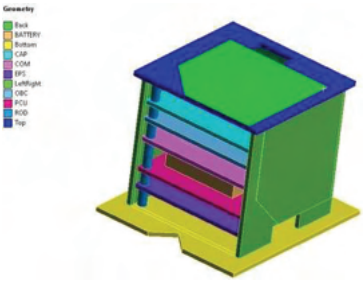
4.3 Thermal-Structural analysis on Satellite



Battery Pack



Results



- The satellite body was consist of a battery pack due to which there was heating issue observed during testing which deforming the materials.
- Therefore, in order to determine the temperature values during it's working in space a transient structural-thermal analysis has been done so as by the limiting results the precautionary cooling arrangements can be done.
- Hence to understand the thermal behavior the transient structural-thermal analysis has been done.

5

PRODUCT ENGINEERING

Our Work



Machine Design



Gold Wash Plant



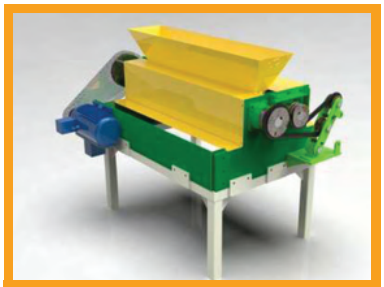
Industrial Blender



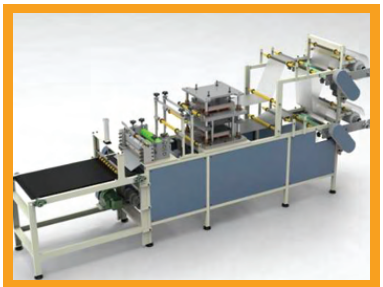
Garbage Cleaner



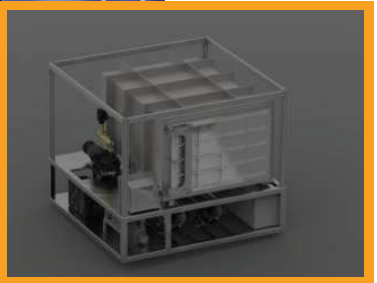
Mixing Auger



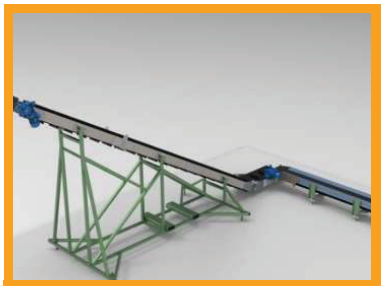
Crumbler (Crusher)



Garlic Cleaning Line



Helium Heat Testing



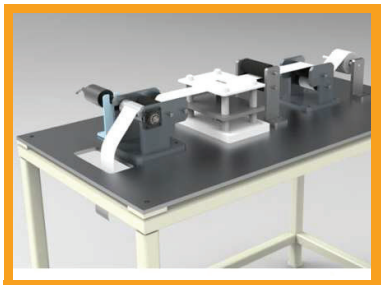
Belt Conveyer



Automated Feed Plant



Scissor Left



Breather Cutting Machine



Flat Belt Conveyor

PRODUCT ENGINEERING
Our Work



Machine Design



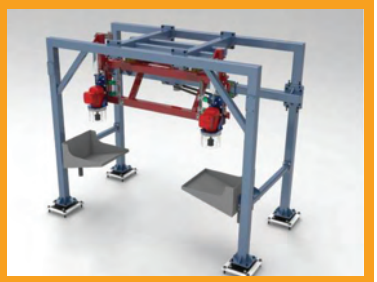
Granite Multiwire Saw



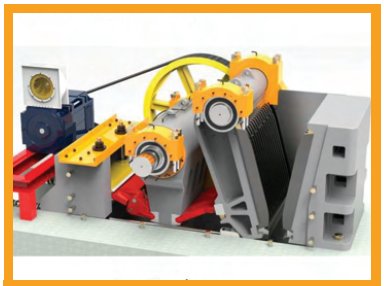
Automotive Conveyor Tech



Lifting Station



Cleaning Station



Saw Crusher



Packing Machine



Buffer Conveyor

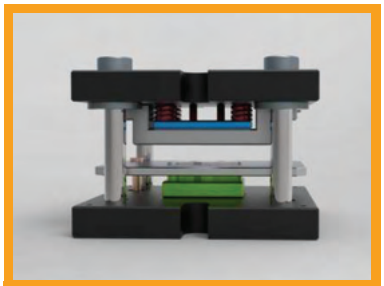


Garlic Cleaning Line

Jig & Fixture, Inspection Gauge, Press Tool Design



DPF Vetri Gauge



Compund Die

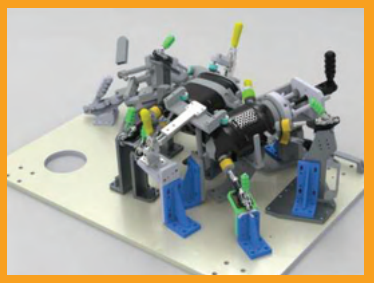


End Gauge

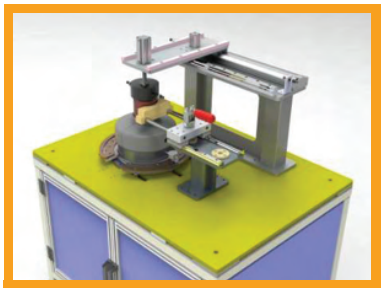
PRODUCT ENGINEERING
Our Work



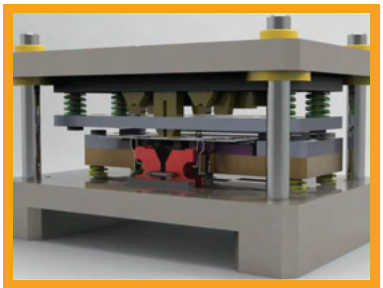
Jig & Fixture, Inspection Gauge, Press Tool Design



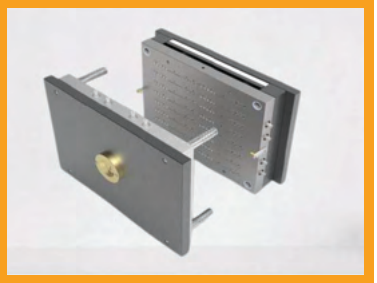
Euro Leak Tester



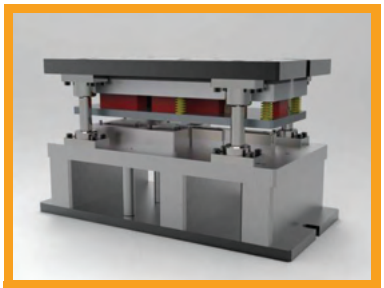
DPF Gauge



Progressive Tool



Strap Mould



Progressive Tool

Product Design



Muffler Design



Smart Wifi
Switch Board



Camera Enclosure



Screw Feeder



Rope Courece



Sliding Robot
for Pipe Weld

PRODUCT ENGINEERING
Our Work



Miscellaneous



Screw Feeder



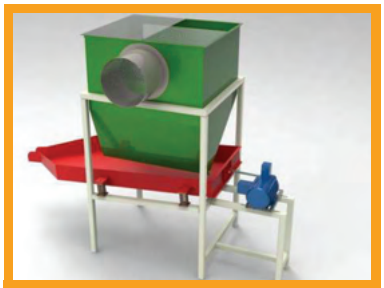
Rope Courece



Sliding Robot for Pipe Weld



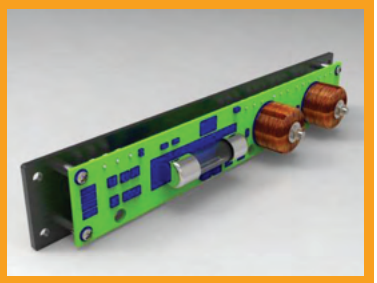
Industrial Shed



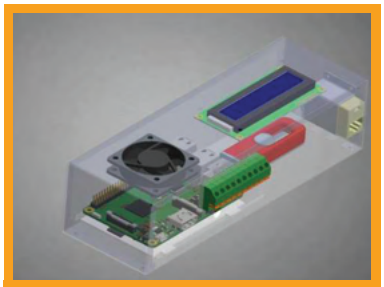
Pellet Cooler



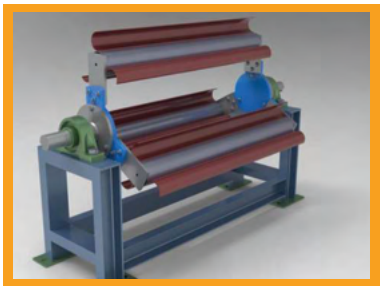
Structure Design



Oil Gas Pressure Sensor



Solar Controller



Turbine



Tool Stand



Pressure Vessel



Chemial Plant



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